HW 2

Bryan Mui - UID 506021334

2025-01-29

Problem 1

1a, create a simulated dataset

simulate(): generate 100 UIDs starting from 100000000 Create homework 1-5 and quiz 1-7, using floor and runif to get a randomized value between 1-100

```
gradebook <- simulate_data()
# arrange by increasing UID
gradebook <- gradebook %>%
  arrange(UID)
head(gradebook)
```

```
## # A tibble: 6 x 13
##         UID Homework_1 Homework_2 Homework_3 Homework_4 Homework_5 Quiz_1 Quiz_2
##       <int>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>  <dbl>  <dbl>
## 1 105467322         80         79        100         91         53     62     78
## 2 107401527         68         62         90         92         55     68     87
## 3 117233698         80         79         79         83         81     83     66
## 4 121937536         82         52         83         78         54     79     68
## 5 124861589         72         65         73         75         84     94     79
## 6 126100703         84         73         50         94         63     71     85
## # i 5 more variables: Quiz_3 <dbl>, Quiz_4 <dbl>, Quiz_5 <dbl>, Quiz_6 <dbl>,
## #   Quiz_7 <dbl>
```

1b) write R code in the R markdown file to randomly replace 10% of the values in HW1, HW5, and Quiz3 with NA values. Use is.na() and sum() to display the results

Algorithm: replace_10_percent() generate random indices as a logical vector(replace or not replace) subset the data and replace TRUE index with NA

```
gradebook$Homework_1 <- replace_10_percent(gradebook$Homework_1)
gradebook$Homework_5 <- replace_10_percent(gradebook$Homework_5)
gradebook$Quiz_3 <- replace_10_percent(gradebook$Quiz_3)

# summary of the replacements:
gradebook %>%
  select(Homework_1, Homework_5, Quiz_3) %>%
  summarise(sum(is.na(.)))
```

```
## # A tibble: 1 x 1
##   `sum(is.na(.))`
##             <int>
## 1              30
```

```r
cat("30 data points should be replaced as NA.")
```

```
## 30 data points should be replaced as NA.
```

```r
# peek the NA values
head(gradebook, 20)
```

```
## # A tibble: 20 x 13
##          UID Homework_1 Homework_2 Homework_3 Homework_4 Homework_5 Quiz_1 Quiz_2
##        <int>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>  <dbl>  <dbl>
##   1   1.05e8         80         79        100         91         53     62     78
##   2   1.07e8         68         62         90         92         NA     68     87
##   3   1.17e8         NA         79         79         83         81     83     66
##   4   1.22e8         82         52         83         78         54     79     68
##   5   1.25e8         72         65         73         75         84     94     79
##   6   1.26e8         84         73         50         94         63     71     85
##   7   1.26e8         91         55         73         65         95     78    100
##   8   1.64e8         64        100         93         65         68     94    100
##   9   2.01e8         85         87         67         94         90     69     90
## 10   2.13e8         84         80         84         76         86     92     53
## 11   2.26e8         NA         81         94         76         55     91     79
## 12   2.30e8         61         75         83         72         82     92     61
## 13   2.44e8         69         89        100         75         62     49     77
## 14   2.46e8         57         42         69         73         88     73     54
## 15   2.62e8         75         62         54         60         52     74     90
## 16   2.83e8         NA         66         79         89         60     55     80
## 17   2.92e8         NA         75         54         78         73     65     73
## 18   2.96e8         66         91         82         92         89    100     72
## 19   3.01e8         97         82         83         51         87     68     84
## 20   3.04e8         94         60         79         72         56     51     58
## # i 5 more variables: Quiz_3 <dbl>, Quiz_4 <dbl>, Quiz_5 <dbl>, Quiz_6 <dbl>,
## #   Quiz_7 <dbl>
```

1c) Write a function messy_impute() that imputes using missing values in the gradebook

algorithm: messy_impute()   set argument defaults   check if center is mean, median, or min; throw error if none of the above   check if margin is 1 or 2   case: columns(margin=1):   for every column in the data frame: summarize by imputing function on each column   case: rows(margin=2):   check the column splits between homeworks and quizes   create a vector of indices subsets for quizzes and homeworks   use filter to select the row and apply the impute function on the quiz set and column set of the data

function implemented in R file

```r
# gradebook[[3]]
head(gradebook, 20)
```

```
## # A tibble: 20 x 13
##          UID Homework_1 Homework_2 Homework_3 Homework_4 Homework_5 Quiz_1 Quiz_2
##        <int>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>  <dbl>  <dbl>
##  1    1.05e8         80         79        100         91         53     62     78
##  2    1.07e8         68         62         90         92         NA     68     87
##  3    1.17e8         NA         79         79         83         81     83     66
##  4    1.22e8         82         52         83         78         54     79     68
##  5    1.25e8         72         65         73         75         84     94     79
##  6    1.26e8         84         73         50         94         63     71     85
##  7    1.26e8         91         55         73         65         95     78    100
##  8    1.64e8         64        100         93         65         68     94    100
##  9    2.01e8         85         87         67         94         90     69     90
## 10    2.13e8         84         80         84         76         86     92     53
## 11    2.26e8         NA         81         94         76         55     91     79
## 12    2.30e8         61         75         83         72         82     92     61
## 13    2.44e8         69         89        100         75         62     49     77
## 14    2.46e8         57         42         69         73         88     73     54
## 15    2.62e8         75         62         54         60         52     74     90
## 16    2.83e8         NA         66         79         89         60     55     80
## 17    2.92e8         NA         75         54         78         73     65     73
## 18    2.96e8         66         91         82         92         89    100     72
## 19    3.01e8         97         82         83         51         87     68     84
## 20    3.04e8         94         60         79         72         56     51     58
## # i 5 more variables: Quiz_3 <dbl>, Quiz_4 <dbl>, Quiz_5 <dbl>, Quiz_6 <dbl>,
## #   Quiz_7 <dbl>
```

```r
head(messy_impute(gradebook, "min", margin=2, trim=0), 20)
```

```
## row  1
## row  2
## row  3
## row  4
## row  5
## row  6
## row  7
## row  8
## row  9
## row  10
## row  11
## row  12
## row  13
## row  14
## row  15
## row  16
## row  17
## row  18
## row  19
## row  20
## row  21
## row  22
## row  23
## row  24
## row  25
## row  26
```

```
## row  27
## row  28
## row  29
## row  30
## row  31
## row  32
## row  33
## row  34
## row  35
## row  36
## row  37
## row  38
## row  39
## row  40
## row  41
## row  42
## row  43
## row  44
## row  45
## row  46
## row  47
## row  48
## row  49
## row  50
## row  51
## row  52
## row  53
## row  54
## row  55
## row  56
## row  57
## row  58
## row  59
## row  60
## row  61
## row  62
## row  63
## row  64
## row  65
## row  66
## row  67
## row  68
## row  69
## row  70
## row  71
## row  72
## row  73
## row  74
## row  75
## row  76
## row  77
## row  78
## row  79
## row  80
```

```
## row  81
## row  82
## row  83
## row  84
## row  85
## row  86
## row  87
## row  88
## row  89
## row  90
## row  91
## row  92
## row  93
## row  94
## row  95
## row  96
## row  97
## row  98
## row  99
## row  100
```

```
## # A tibble: 20 x 13
##          UID Homework_1 Homework_2 Homework_3 Homework_4 Homework_5 Quiz_1 Quiz_2
##        <int>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>  <dbl>  <dbl>
##  1    1.05e8         80         79        100         91         53     62     78
##  2    1.07e8         68         62         90         92         62     68     87
##  3    1.17e8         79         79         79         83         81     83     66
##  4    1.22e8         82         52         83         78         54     79     68
##  5    1.25e8         72         65         73         75         84     94     79
##  6    1.26e8         84         73         50         94         63     71     85
##  7    1.26e8         91         55         73         65         95     78    100
##  8    1.64e8         64        100         93         65         68     94    100
##  9    2.01e8         85         87         67         94         90     69     90
## 10    2.13e8         84         80         84         76         86     92     53
## 11    2.26e8         55         81         94         76         55     91     79
## 12    2.30e8         61         75         83         72         82     92     61
## 13    2.44e8         69         89        100         75         62     49     77
## 14    2.46e8         57         42         69         73         88     73     54
## 15    2.62e8         75         62         54         60         52     74     90
## 16    2.83e8         60         66         79         89         60     55     80
## 17    2.92e8         54         75         54         78         73     65     73
## 18    2.96e8         66         91         82         92         89    100     72
## 19    3.01e8         97         82         83         51         87     68     84
## 20    3.04e8         94         60         79         72         56     51     58
## # i 5 more variables: Quiz_3 <dbl>, Quiz_4 <dbl>, Quiz_5 <dbl>, Quiz_6 <dbl>,
## #   Quiz_7 <dbl>
```

1d) Please select one student whose quiz or homework has undergone imputation to showcase your function

```r
# Row 2 has a NA value
head(messy_impute(gradebook, "mean", 1))
```

```
## # A tibble: 6 x 13
##         UID Homework_1 Homework_2 Homework_3 Homework_4 Homework_5 Quiz_1 Quiz_2
##       <int>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>  <dbl>  <dbl>
## 1 105467322       80           79        100         91       53       62     78
## 2 107401527       68           62         90         92       75.4     68     87
## 3 117233698       73.8         79         79         83       81       83     66
## 4 121937536       82           52         83         78       54       79     68
## 5 124861589       72           65         73         75       84       94     79
## 6 126100703       84           73         50         94       63       71     85
## # i 5 more variables: Quiz_3 <dbl>, Quiz_4 <dbl>, Quiz_5 <dbl>, Quiz_6 <dbl>,
## #   Quiz_7 <dbl>
```

```r
head(messy_impute(gradebook, "median", 2))
```

```
## row  1
## row  2
## row  3
## row  4
## row  5
## row  6
## row  7
## row  8
## row  9
## row  10
## row  11
## row  12
## row  13
## row  14
## row  15
## row  16
## row  17
## row  18
## row  19
## row  20
## row  21
## row  22
## row  23
## row  24
## row  25
## row  26
## row  27
## row  28
## row  29
## row  30
## row  31
## row  32
## row  33
## row  34
## row  35
## row  36
## row  37
## row  38
## row  39
## row  40
```

```
## row  41
## row  42
## row  43
## row  44
## row  45
## row  46
## row  47
## row  48
## row  49
## row  50
## row  51
## row  52
## row  53
## row  54
## row  55
## row  56
## row  57
## row  58
## row  59
## row  60
## row  61
## row  62
## row  63
## row  64
## row  65
## row  66
## row  67
## row  68
## row  69
## row  70
## row  71
## row  72
## row  73
## row  74
## row  75
## row  76
## row  77
## row  78
## row  79
## row  80
## row  81
## row  82
## row  83
## row  84
## row  85
## row  86
## row  87
## row  88
## row  89
## row  90
## row  91
## row  92
## row  93
## row  94
```

```
## row  95
## row  96
## row  97
## row  98
## row  99
## row  100
```

```
## # A tibble: 6 x 13
##         UID Homework_1 Homework_2 Homework_3 Homework_4 Homework_5 Quiz_1 Quiz_2
##       <int>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>  <dbl>  <dbl>
## 1 105467322         80         79        100         91         53     62     78
## 2 107401527         68         62         90         92         79     68     87
## 3 117233698         80         79         79         83         81     83     66
## 4 121937536         82         52         83         78         54     79     68
## 5 124861589         72         65         73         75         84     94     79
## 6 126100703         84         73         50         94         63     71     85
## # i 5 more variables: Quiz_3 <dbl>, Quiz_4 <dbl>, Quiz_5 <dbl>, Quiz_6 <dbl>,
## #   Quiz_7 <dbl>
```

1e) Write R code using the main functions in the tidyr package to convert the gradebook into the tidy format

```r
# Convert gradebook to tidy
gradebook_tidy <- gradebook %>%
  pivot_longer(
    cols = -UID,
    names_to = "Assesment",
    values_to = "Score"
  ) %>%
  separate(
    col = Assesment,
    into = c("Assesment", "Number"),
    sep = "_"
  ) %>%
  mutate(Number=as.numeric(.$Number))
head(gradebook_tidy, 15)
```

```
## # A tibble: 15 x 4
##          UID Assesment Number Score
##        <int> <chr>      <dbl> <dbl>
##   1 105467322 Homework       1    80
##   2 105467322 Homework       2    79
##   3 105467322 Homework       3   100
##   4 105467322 Homework       4    91
##   5 105467322 Homework       5    53
##   6 105467322 Quiz           1    62
##   7 105467322 Quiz           2    78
##   8 105467322 Quiz           3    63
##   9 105467322 Quiz           4    67
##  10 105467322 Quiz           5    84
##  11 105467322 Quiz           6    48
```

```
## 12 105467322 Quiz          7    77
## 13 107401527 Homework      1    68
## 14 107401527 Homework      2    62
## 15 107401527 Homework      3    90
```

1f) Write a function tidy impute() that imputes missing values in gradebook tidy object

algorithm: tidy_impute(): set argument defaults check if center is mean, median, or min; throw error if none of the above check if margin is 1 or 2 find distinct uid names case: columns(margin=1): for every uid for that assignment(subsetted from distinct uid names) impute on the correct function case: rows(margin=2): select rows with filter for correct UID and homework/quizzes apply the impute function on the quiz set and column set of the data

function declared in R script

```r
tidy_impute(gradebook_tidy, "mean", margin=1, trim=0)
```

```
## # A tibble: 1,200 x 4
##          UID Assesment Number Score
##        <int> <chr>      <dbl> <list>
##  1 105467322 Homework       1 <dbl [1]>
##  2 105467322 Homework       2 <dbl [1]>
##  3 105467322 Homework       3 <dbl [1]>
##  4 105467322 Homework       4 <dbl [1]>
##  5 105467322 Homework       5 <dbl [1]>
##  6 105467322 Quiz           1 <dbl [1]>
##  7 105467322 Quiz           2 <dbl [1]>
##  8 105467322 Quiz           3 <dbl [1]>
##  9 105467322 Quiz           4 <dbl [1]>
## 10 105467322 Quiz           5 <dbl [1]>
## # i 1,190 more rows
```

1g) Please use the cases you select from (d) to demonstrate your function in the R markdown

```r
head(tidy_impute(gradebook_tidy, "mean", 1))
```

```
## Warning: There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
```

```
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
```

```
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
```

```
## # A tibble: 6 x 4
##         UID Assesment Number Score
##       <int> <chr>      <dbl> <list>
## 1 105467322 Homework       1 <dbl [1]>
## 2 105467322 Homework       2 <dbl [1]>
## 3 105467322 Homework       3 <dbl [1]>
## 4 105467322 Homework       4 <dbl [1]>
## 5 105467322 Homework       5 <dbl [1]>
## 6 105467322 Quiz           1 <dbl [1]>
```

```r
head(tidy_impute(gradebook_tidy, "median", 2))
```

```
## Warning: There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
```

```
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
```

```
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## There was 1 warning in ‘summarize()‘.
## i In argument: ‘mean(., na.rm = TRUE)‘.
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
```

```
## # A tibble: 6 x 4
##         UID Assesment Number Score
##       <int> <chr>      <dbl> <list>
## 1 105467322 Homework       1 <dbl [1]>
## 2 105467322 Homework       2 <dbl [1]>
## 3 105467322 Homework       3 <dbl [1]>
## 4 105467322 Homework       4 <dbl [1]>
## 5 105467322 Homework       5 <dbl [1]>
## 6 105467322 Quiz           1 <dbl [1]>
```

Section 2

1a find 3 examples of messy data

one example would be a beauty ratings conditions data set, that has reviewers and the columns with the various subjects so the data is not tidy because the multiple reviews of all the subjects are on one row. Another would be election data, which would show a table with the election results for democrats/republicans and the columns would be each year, and each data point would be the election votes. This would be easier to visualize but is not tidy, because there are multiple observations per row. One final example might be the gradebook in the homework. Each row is a UID, and assignments and quizes are in the columns. Data would be stored this way so that it is easy to look up each student and then see their grades for all the exams. However, there are multiple scores per row, which would mean that there is more than one observation per row in the example.